NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-28776/4E
AMENDMENT 4
29 October 1992
SUPERSEDING
AMENDMENT 3
1 July 1992

PERFORMANCE SPECIFICATION SHEET

RELAYS, HYBRID, ESTABLISHED RELIABILITY, SPDT, LOW LEVEL TO 1.0 AMPERE (SENSITIVE, COIL OPERATE POWER AT 25°C) (TRANSISTOR DRIVEN) (ELECTROMECHANICAL OUTPUT) WITH DIODE COIL SUPPRESSION

This amendment forms a part of MIL-R-28776/4E, dated 10 February 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

- TABLE I, delete in its entirety and substitute new table I as printed on page 2 of this amendment.

PAGE 4

REQUIREMENTS, Resistive, delete in its entirety and substitute:

"Resistive:
1.0 ampere at 28 V dc.
250 milliamperes at 115 V ac 60 and 400 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded."

REQUIREMENTS, Overload (high level only), delete and substitute:

"Overload (high level only): Two times rated current. Not applicable to ac load ratings."

PAGE 5

ENVIRONMENTAL DATA, delete "Shock" and "Vibration" in their entirety and substitute:

"Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

"Vibration (sinusoidal): MIL-STD-202, method 204, (30 g's, 10 to 3,000 Hz). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts."

PAGE 7

- Qualification inspection: Delete "Group A testing for relays supplied with spreader pads (-041 through 048), shall be tested as specified below:" and substitute:

"Perform seal test immediately preceding the electrical tests. Relay leads shall be formed and the mounting pad removed before the seal test. After the seal test, the mounting pad shall be rigidly attached to the relay and the remaining group A tests performed."

AMSC N/A [illegible] FSC 5945

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/				Supply voltage (V dc) 5/		At +25°C							Over temperature range			
Lead length			Spreader pad (fig 2) 4/	Rated	Max	Coil resistance (ohms) (±10%) 6/	Coil circuit current mA 6/ 7/		Pick-voltage (V dc) Max 7/	Hold voltage (V dc) 7/	Drop-out voltage (V dc) 7/	Max base current to assure turn on (mA)	Pick-up voltage (V dc) Max 7/	Hold voltage (V dc) 7/	Drop-out voltage (V dc) 7/	Max base current to assure turn on (mA)
1.500 min 3/	.187 +.050, -.010	.500 min					Max	Min								
017	025	033	041	5.0	8.0	125	47.8	34.7	2.6	1.4	0.22	.28	3.6	2.0	0.15	1.20
018	026	034	042	6.0	11.0	255	27.7	21.2	3.5	2.0	0.28	.20	4.8	2.8	0.18	0.78
019	027	035	043	9.0	16.0	630	16.8	11.8	5.4	3.0	0.54	.13	7.8	4.2	0.35	0.48
020	028	036	044	12.0	22.0	1,025	13.6	10.1	6.6	4.0	0.63	.10	10.0	5.6	0.41	0.39
021	029	037	045	18.0	33.0	2,300	9.1	6.7	9.8	6.0	0.91	.07	14.5	8.4	0.58	0.26
022	030	038	046	26.5	45.0	4,000	7.7	5.7	12.8	8.0	1.37	.05	19.0	10.4	0.89	0.20
023	031	039	047	32.0	57.0	6,500	5.8	4.2	18.7	10.0	1.60	.04	24.0	15.0	1.00	0.16
024	032	040	048	40.0	75.0	11,000	4.3	3.1	23.3	13.0	2.10	.03	30.0	18.7	1.30	0.13

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01.
Example 017L - - - - - 032R.

3/ 1.500 leads are inactive for new design.

4/ Relays supplied with spreader pads (-041 through -048) shall have the pad rigidly attached.

5/ CAUTION: The use of any supply voltage less than the rated supply voltage will compromise the operation of the relay.

6/ Coil resistance not directly measurable at relay terminals. When rated supply voltage is applied to the supply terminals, the coil circuit current must be within the limits shown. Measure at 25°C at rated supply voltage for 5 seconds, maximum.

7/ Set base current from 3 mA to 15 mA during electrical measurements.

The margins of this amendment are marked with an asterisk (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
  Army - ER
  Navy - EC
  Air Force - 85

Review activities:
  Army - MI
  Air Force - 17, 99
  DLA - ES
  NSA - NS

User activities:
  Army - AR, ME
  Navy - AS, OS, SH
  Air Force - 19

Preparing activity:
  Navy - EC

Agent:
  DLA - ES

(Project 5945-0899-03)